

gmub9mvnr

November 7, 2025

1 Exel

```
[1]: import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Flatten, Dense, Rescaling
from tensorflow.keras.utils import image_dataset_from_directory
```

```
C:\Users\admin\AppData\Roaming\Python\Python311\site-
packages\pandas\core\arrays\masked.py:60: UserWarning: Pandas requires version
'1.3.6' or newer of 'bottleneck' (version '1.3.5' currently installed).
from pandas.core import (
```

```
[23]: train_ds=image_dataset_from_directory(
        'E:/126156040/Vegetable Images/train',
        image_size=(64,64),batch_size=32,label_mode='categorical')

val_ds=image_dataset_from_directory(
        'E:/126156040/Vegetable Images/validation',
        image_size=(64,64),batch_size=32,label_mode='categorical')

test_ds=image_dataset_from_directory(
        'E:/126156040/Vegetable Images/test',
        image_size=(64,64),batch_size=32,label_mode='categorical')
```

```
Found 15000 files belonging to 15 classes.
Found 3000 files belonging to 15 classes.
Found 3000 files belonging to 15 classes.
```

```
[24]: normalization_layer=Rescaling(1./255)
train_ds=train_ds.map(lambda x,y: (normalization_layer(x),y))
val_ds=val_ds.map(lambda x,y: (normalization_layer(x),y))
test_ds=test_ds.map(lambda x,y: (normalization_layer(x),y))
```

```
[25]: model = Sequential([
        Flatten(input_shape=(64, 64, 3)),
        Dense(128, activation='relu'),
        Dense(64, activation='relu'),
        Dense(15, activation='softmax')
    ])
```

C:\Users\admin\AppData\Roaming\Python\Python311\site-packages\keras\src\layers\reshaping\flatten.py:37: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.
super().__init__(**kwargs)

```
[26]: model.compile(optimizer='adam',
                    loss='categorical_crossentropy',
                    metrics=['accuracy'])
```

```
[27]: history=model.fit(train_ds, validation_data=val_ds,epochs=5)
```

```
Epoch 1/5
469/469          14s 28ms/step -
accuracy: 0.2593 - loss: 2.4454 - val_accuracy: 0.4760 - val_loss: 1.6462
Epoch 2/5
469/469          12s 27ms/step -
accuracy: 0.4803 - loss: 1.5971 - val_accuracy: 0.5900 - val_loss: 1.3393
Epoch 3/5
469/469          12s 26ms/step -
accuracy: 0.5779 - loss: 1.3100 - val_accuracy: 0.6223 - val_loss: 1.1962
Epoch 4/5
469/469          12s 25ms/step -
accuracy: 0.6376 - loss: 1.1276 - val_accuracy: 0.6433 - val_loss: 1.1197
Epoch 5/5
469/469          13s 27ms/step -
accuracy: 0.6786 - loss: 1.0180 - val_accuracy: 0.6653 - val_loss: 1.0565
```

```
[28]: loss,acc = model.evaluate(test_ds)
print(f'Test Accuracy: {acc:.4f}, Test Loss: {loss:.4f}')
```

```
94/94           1s 8ms/step -
accuracy: 0.6898 - loss: 0.9903
Test Accuracy: 0.6750, Test Loss: 1.0250
```

```
[29]: plt.figure(figsize=(12,5))
plt.subplot(1,2,1)
plt.plot(history.history['accuracy'],label='Train Acc')
plt.plot(history.history['val_accuracy'],label='Val Acc')
plt.title('Accuracy')
```

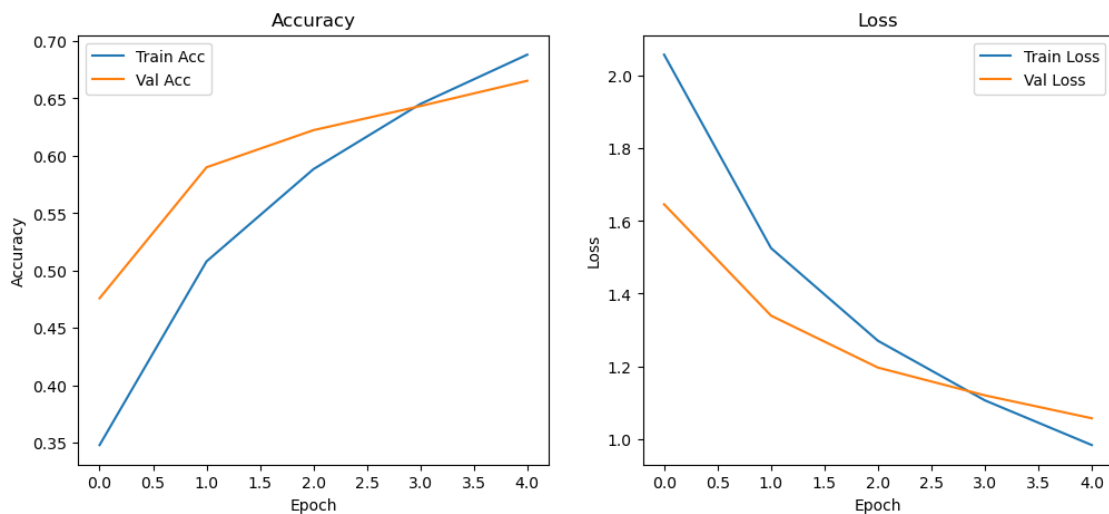
```

plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend()

plt.subplot(1,2,2)
plt.plot(history.history['loss'],label='Train Loss')
plt.plot(history.history['val_loss'],label='Val Loss')
plt.title('Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()

plt.show()

```



```

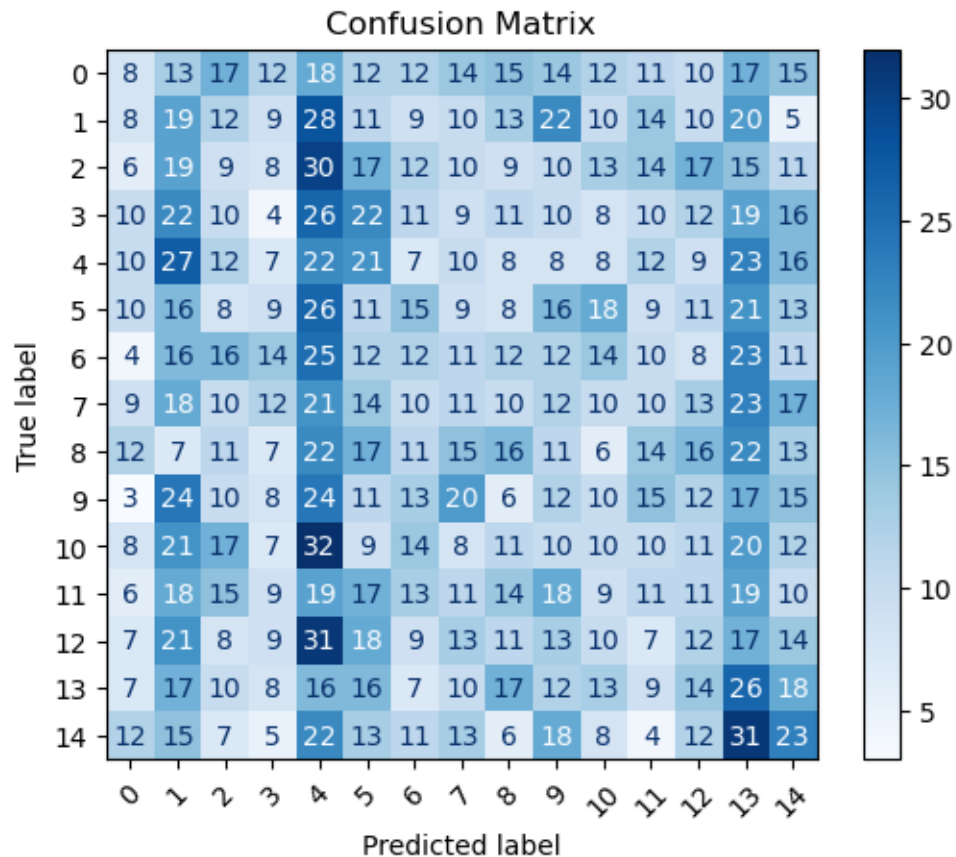
[30]: y_true=np.concatenate([y for x,y in test_ds],axis=0)
      y_pred_prob=model.predict(test_ds)
      y_pred=np.argmax(y_pred_prob,axis=1)
      y_true_label=np.argmax(y_true,axis=1)

```

WARNING:tensorflow:5 out of the last 15 calls to <function TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_distributed at 0x000001CCB9568360> triggered tf.function retracing. Tracing is expensive and the excessive number of tracings could be due to (1) creating @tf.function repeatedly in a loop, (2) passing tensors with different shapes, (3) passing Python objects instead of tensors. For (1), please define your @tf.function outside of the loop. For (2), @tf.function has reduce_retracing=True option that can avoid unnecessary retracing. For (3), please refer to https://www.tensorflow.org/guide/function#controlling_retracing and https://www.tensorflow.org/api_docs/python/tf/function for more details.

94/94 1s 6ms/step

```
[31]: cm=confusion_matrix(y_true_label,y_pred)
disp=ConfusionMatrixDisplay(cm)
disp.plot(cmap=plt.cm.Blues, xticks_rotation=45)
plt.title('Confusion Matrix')
plt.show()
```



2 Exe 2

```
[2]: import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow.keras.utils import image_dataset_from_directory
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, Flatten, MaxPooling2D, Dense,
↳Dropout, Rescaling
from sklearn.metrics import classification_report, confusion_matrix,
↳ConfusionMatrixDisplay
```

```
[5]: dataset_path='E:/126156040/Bean_Dataset'
      batch_size=32
      img_size=(128,128)
```

```
[6]: train_ds = image_dataset_from_directory(
      dataset_path, validation_split=0.2,
      subset="training", seed=42,
      image_size=img_size, batch_size=batch_size)

      val_ds = image_dataset_from_directory(
      dataset_path, validation_split=0.2,
      subset="validation", seed=42,
      image_size=img_size, batch_size=batch_size)
```

Found 990 files belonging to 3 classes.
Using 792 files for training.
Found 990 files belonging to 3 classes.
Using 198 files for validation.

```
[7]: class_names=train_ds.class_names
      num_classes=len(class_names)

      train_ds=train_ds.map(lambda x,y: (x/255,y)).cache().prefetch(tf.data.AUTOTUNE)
      val_ds=val_ds.map(lambda x,y: (x/255,y)).cache().prefetch(tf.data.AUTOTUNE)
```

```
[8]: def flatten_dataset(ds):
      x,y=[],[]
      for images,labels in ds:
          x.append(tf.reshape(images,(images.shape[0],-1)))
          y.append(labels)
      return tf.concat(x,axis=0), tf.concat(y,axis=0)
      x_train_flat, y_train= flatten_dataset(train_ds)
      x_val_flat, y_val=flatten_dataset(val_ds)
```

```
[9]: dnn=Sequential([
      Dense(26,activation='relu',input_shape=(x_train_flat.shape[1],)),
      Dense(128,activation='relu'),
      Dense(num_classes,activation='softmax')
      ])
```

C:\Users\admin\AppData\Roaming\Python\Python311\site-packages\keras\src\layers\core\dense.py:87: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.
super().__init__(activity_regularizer=activity_regularizer, **kwargs)

```
[10]: dnn.
      ↪ compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])
dnn_history = dnn.fit(x_train_flat, y_train, validation_data=(x_val_flat,
      ↪ y_val), epochs=10, batch_size=32)
```

```
Epoch 1/10
25/25          1s 21ms/step -
accuracy: 0.3988 - loss: 2.8614 - val_accuracy: 0.5253 - val_loss: 1.5174
Epoch 2/10
25/25          0s 16ms/step -
accuracy: 0.5341 - loss: 1.0846 - val_accuracy: 0.5758 - val_loss: 0.8890
Epoch 3/10
25/25          0s 18ms/step -
accuracy: 0.6339 - loss: 0.8450 - val_accuracy: 0.5859 - val_loss: 0.9379
Epoch 4/10
25/25          0s 17ms/step -
accuracy: 0.6314 - loss: 0.8418 - val_accuracy: 0.6414 - val_loss: 0.7877
Epoch 5/10
25/25          0s 16ms/step -
accuracy: 0.6949 - loss: 0.7002 - val_accuracy: 0.6061 - val_loss: 0.9883
Epoch 6/10
25/25          0s 16ms/step -
accuracy: 0.7525 - loss: 0.6286 - val_accuracy: 0.5556 - val_loss: 1.0029
Epoch 7/10
25/25          0s 16ms/step -
accuracy: 0.6005 - loss: 0.8515 - val_accuracy: 0.6414 - val_loss: 0.7990
Epoch 8/10
25/25          0s 16ms/step -
accuracy: 0.7823 - loss: 0.5335 - val_accuracy: 0.6768 - val_loss: 0.7638
Epoch 9/10
25/25          0s 16ms/step -
accuracy: 0.7799 - loss: 0.5501 - val_accuracy: 0.5808 - val_loss: 1.0209
Epoch 10/10
25/25          0s 16ms/step -
accuracy: 0.6791 - loss: 0.7126 - val_accuracy: 0.5909 - val_loss: 1.2767
```

```
[11]: dnn_loss, dnn_acc = dnn.evaluate(x_val_flat, y_val)
      print(f'DNN Accuracy: {dnn_acc*100:.2f}%')
      print(f'DNN Parameters: {dnn.count_params()}')
```

```
7/7           0s 2ms/step -
accuracy: 0.5921 - loss: 1.3354
DNN Accuracy: 59.09%
DNN Parameters: 1281821
```

```
[18]: cnn = Sequential([
```

```

    Conv2D(32, (3,3),  

    ↪activation='relu',input_shape=(img_size[0],img_size[1],3)),  

    MaxPooling2D(2,2),  

    Conv2D(64, (3,3), activation='relu'),  

    MaxPooling2D(2,2),  

    Flatten(),  

    Dense(128, activation='relu'),  

    Dropout(0.5),  

    Dense(num_classes,activation='softmax')  

])

```

```

[19]: cnn.compile(optimizer='adam', loss='sparse_categorical_crossentropy',  

    ↪metrics=['accuracy'])  

cnn_history= cnn.fit(train_ds,validation_data=val_ds,epochs=10)

```

```

Epoch 1/10
25/25          6s 212ms/step -
accuracy: 0.3236 - loss: 1.7334 - val_accuracy: 0.5758 - val_loss: 0.9597
Epoch 2/10
25/25          5s 206ms/step -
accuracy: 0.5347 - loss: 0.9664 - val_accuracy: 0.6263 - val_loss: 0.7709
Epoch 3/10
25/25          5s 208ms/step -
accuracy: 0.6423 - loss: 0.7961 - val_accuracy: 0.6364 - val_loss: 0.7971
Epoch 4/10
25/25          5s 207ms/step -
accuracy: 0.7133 - loss: 0.7181 - val_accuracy: 0.7273 - val_loss: 0.6256
Epoch 5/10
25/25          5s 208ms/step -
accuracy: 0.7677 - loss: 0.5503 - val_accuracy: 0.7071 - val_loss: 0.7019
Epoch 6/10
25/25          5s 207ms/step -
accuracy: 0.8250 - loss: 0.5076 - val_accuracy: 0.7222 - val_loss: 0.5923
Epoch 7/10
25/25          5s 208ms/step -
accuracy: 0.8628 - loss: 0.3686 - val_accuracy: 0.7172 - val_loss: 0.6786
Epoch 8/10
25/25          5s 213ms/step -
accuracy: 0.9056 - loss: 0.2799 - val_accuracy: 0.6818 - val_loss: 0.9090
Epoch 9/10
25/25          5s 215ms/step -
accuracy: 0.8978 - loss: 0.2918 - val_accuracy: 0.7323 - val_loss: 0.7005
Epoch 10/10
25/25          6s 225ms/step -
accuracy: 0.9211 - loss: 0.2203 - val_accuracy: 0.7222 - val_loss: 0.7161

```

```
[20]: cnn_loss,cnn_acc=cnn.evaluate(val_ds)
print(f'CNN Test Accuracy: {cnn_acc*100:.2f}%')
print(f'CNN Parameters: {cnn.count_params()}')
```

```
7/7          0s 34ms/step -
accuracy: 0.7028 - loss: 0.8444
CNN Test Accuracy: 72.22%
CNN Parameters: 7392707
```

```
[21]: def show_cm(model,x,y=None,flat=False):
    if flat:
        y_true=y.numpy()
        y_pred=np.argmax(model.predict(x),axis=1)
    else:
        y_true=np.concatenate([labels.numpy() for _, labels in x],axis=0)
        y_pred=np.argmax(model.predict(x),axis=1)
    print('Classification report:\n',
    ↪',classification_report(y_true,y_pred,target_names=class_names,zero_division=0))
    print('Confusion Matrix:\n',
    ↪ConfusionMatrixDisplay(confusion_matrix(y_true,y_pred),display_labels=class_names).
    ↪plot())

print("DNN Performance:\n",show_cm(dnn,x_val_flat,y_val,flat=True))
print("CNN Performance:\n",show_cm(cnn,val_ds))
```

```
7/7          0s 6ms/step
Classification report:
              precision    recall  f1-score   support

angular_leaf_spot      0.50      0.83      0.63         64
    bean_rust           1.00      0.02      0.03         65
    healthy             0.68      0.91      0.78         69

    accuracy                   0.59         198
    macro avg              0.73      0.59      0.48         198
    weighted avg           0.73      0.59      0.49         198
```

```
Confusion Matrix:
<sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay object at
0x000001CCAF86A510>
```

```
DNN Performance:
None
```

```
7/7          0s 40ms/step
Classification report:
              precision    recall  f1-score   support

angular_leaf_spot      0.69      0.64      0.67         64
```

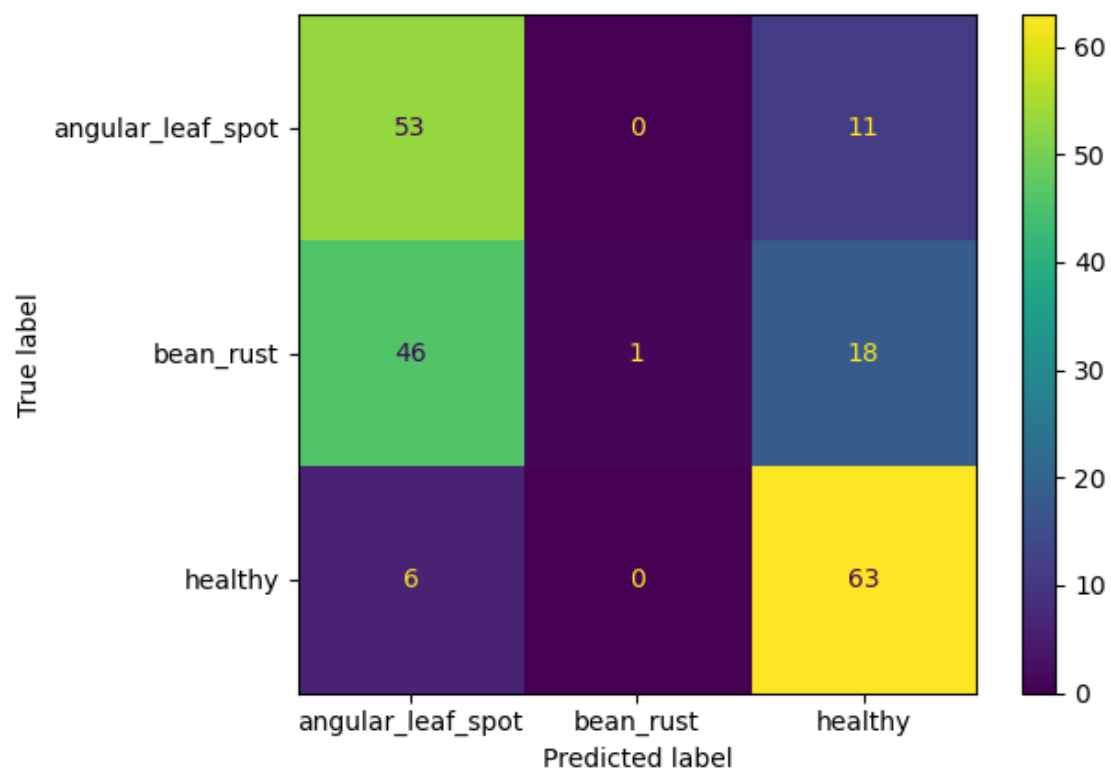

bean_rust	0.76	0.58	0.66	65
healthy	0.72	0.93	0.81	69
accuracy			0.72	198
macro avg	0.72	0.72	0.71	198
weighted avg	0.72	0.72	0.71	198

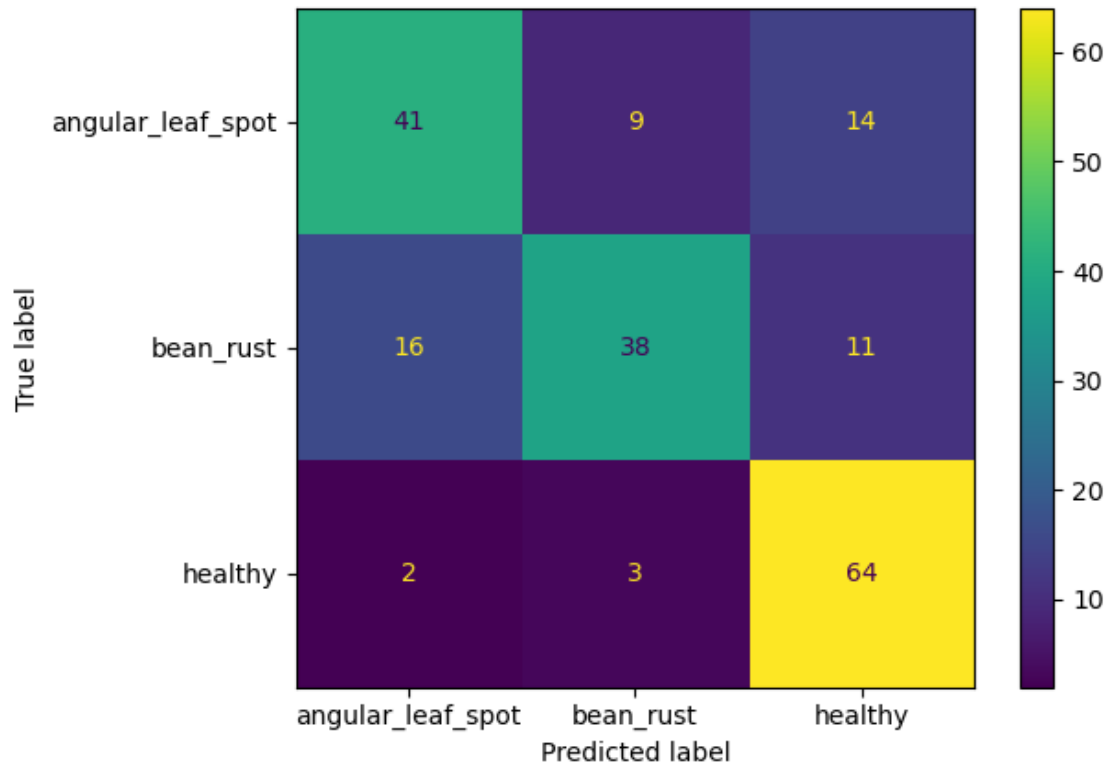
Confusion Matrix:

<sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay object at 0x000001CCB3EF07D0>

CNN Performance:

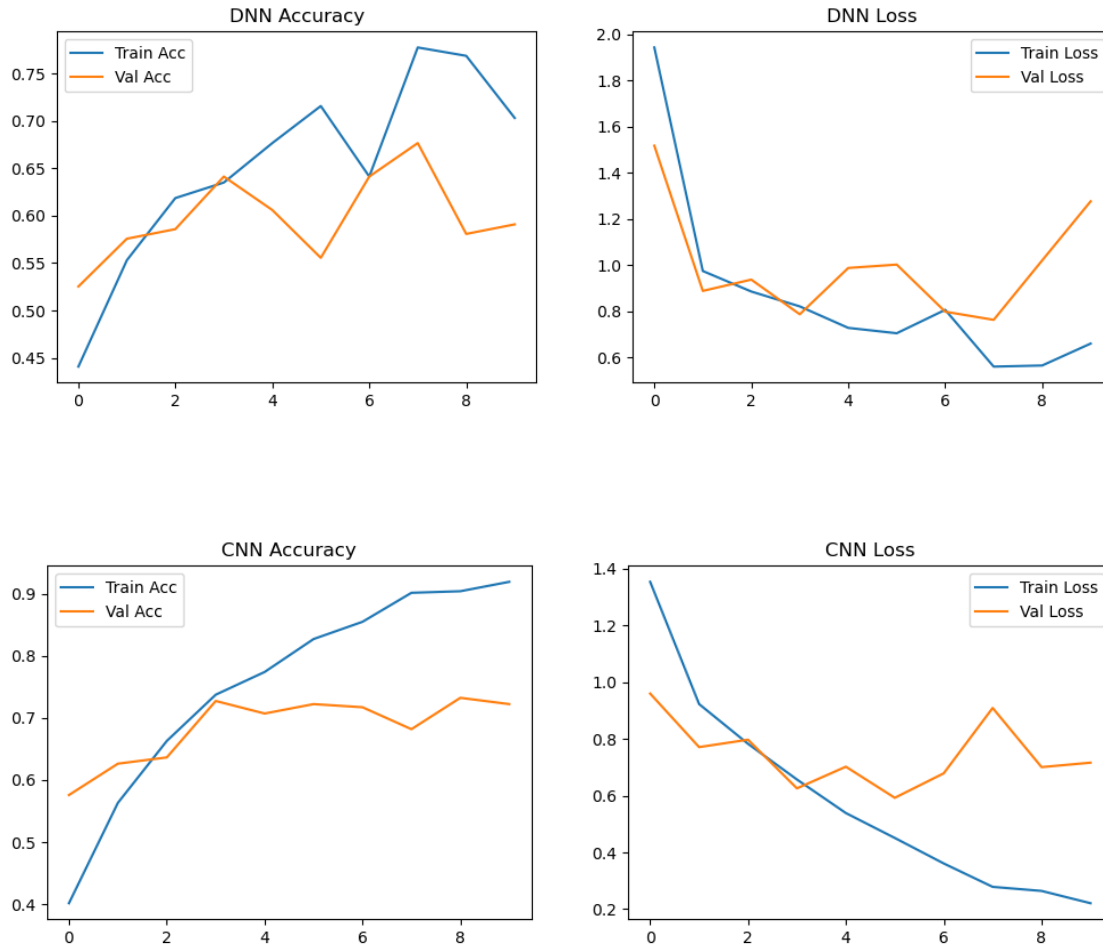
None





```
[22]: def plot_hist(hist, title):
plt.figure(figsize=(12,4))
plt.subplot(1,2,1)
plt.plot(hist.history['accuracy'], label='Train Acc')
plt.plot(hist.history['val_accuracy'], label='Val Acc')
plt.title(f'{title} Accuracy')
plt.legend()
plt.subplot(1,2,2)
plt.plot(hist.history['loss'], label='Train Loss')
plt.plot(hist.history['val_loss'], label='Val Loss')
plt.title(f'{title} Loss')
plt.legend()
plt.show()

plot_hist(dnn_history, "DNN")
plot_hist(cnn_history, "CNN")
```



3 Exe3

```
[52]: !pip install ultralytics
```

```
Defaulting to user installation because normal site-packages is not writeable
Collecting ultralytics
```

```
Obtaining dependency information for ultralytics from https://files.pythonhost
ed.org/packages/a1/78/ea1826897a4beea548b5f79730c627b16a98f00f1843978966fb2c6a5c
c6/ultralytics-8.3.197-py3-none-any.whl.metadata
```

```
Downloading ultralytics-8.3.197-py3-none-any.whl.metadata (37 kB)
```

```
Requirement already satisfied: numpy>=1.23.0 in
c:\users\admin\appdata\roaming\python\python311\site-packages (from ultralytics)
(1.26.4)
```

```
Requirement already satisfied: matplotlib>=3.3.0 in
c:\programdata\anaconda3\lib\site-packages (from ultralytics) (3.7.1)
```

```
Requirement already satisfied: opencv-python>=4.6.0 in
c:\users\admin\appdata\roaming\python\python311\site-packages (from ultralytics)
```

```

(4.9.0.80)
Requirement already satisfied: pillow>=7.1.2 in
c:\programdata\anaconda3\lib\site-packages (from ultralytics) (9.4.0)
Requirement already satisfied: pyyaml>=5.3.1 in
c:\programdata\anaconda3\lib\site-packages (from ultralytics) (6.0)
Requirement already satisfied: requests>=2.23.0 in
c:\programdata\anaconda3\lib\site-packages (from ultralytics) (2.31.0)
Requirement already satisfied: scipy>=1.4.1 in
c:\users\admin\appdata\roaming\python\python311\site-packages (from ultralytics)
(1.12.0)
Requirement already satisfied: torch>=1.8.0 in
c:\users\admin\appdata\roaming\python\python311\site-packages (from ultralytics)
(2.7.1)
Requirement already satisfied: torchvision>=0.9.0 in
c:\users\admin\appdata\roaming\python\python311\site-packages (from ultralytics)
(0.22.1)
Requirement already satisfied: psutil in c:\programdata\anaconda3\lib\site-
packages (from ultralytics) (5.9.0)
Collecting polars (from ultralytics)
  Obtaining dependency information for polars from https://files.pythonhosted.or
g/packages/06/a6/dc535da476c93b2efac619e04ab81081e004e4b4553352cd10e0d33a015d/po
lars-1.33.1-cp39-abi3-win_amd64.whl.metadata
  Downloading polars-1.33.1-cp39-abi3-win_amd64.whl.metadata (15 kB)
Collecting ultralytics-thop>=2.0.0 (from ultralytics)
  Obtaining dependency information for ultralytics-thop>=2.0.0 from https://file
s.pythonhosted.org/packages/b7/02/de5b1f012a9eb319fdd724049902d7b6886b694c281059
e5d734296b74af/ultralytics_thop-2.0.17-py3-none-any.whl.metadata
  Downloading ultralytics_thop-2.0.17-py3-none-any.whl.metadata (14 kB)
Requirement already satisfied: contourpy>=1.0.1 in
c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.3.0->ultralytics)
(1.0.5)
Requirement already satisfied: cycler>=0.10 in
c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.3.0->ultralytics)
(0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in
c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.3.0->ultralytics)
(4.25.0)
Requirement already satisfied: kiwisolver>=1.0.1 in
c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.3.0->ultralytics)
(1.4.4)
Requirement already satisfied: packaging>=20.0 in
c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.3.0->ultralytics)
(23.0)
Requirement already satisfied: pyparsing>=2.3.1 in
c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.3.0->ultralytics)
(3.0.9)
Requirement already satisfied: python-dateutil>=2.7 in
c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.3.0->ultralytics)

```

```

(2.8.2)
Requirement already satisfied: charset-normalizer<4,>=2 in
c:\programdata\anaconda3\lib\site-packages (from requests>=2.23.0->ultralitics)
(2.0.4)
Requirement already satisfied: idna<4,>=2.5 in
c:\programdata\anaconda3\lib\site-packages (from requests>=2.23.0->ultralitics)
(3.4)
Requirement already satisfied: urllib3<3,>=1.21.1 in
c:\programdata\anaconda3\lib\site-packages (from requests>=2.23.0->ultralitics)
(1.26.16)
Requirement already satisfied: certifi>=2017.4.17 in
c:\programdata\anaconda3\lib\site-packages (from requests>=2.23.0->ultralitics)
(2023.7.22)
Requirement already satisfied: filelock in c:\programdata\anaconda3\lib\site-
packages (from torch>=1.8.0->ultralitics) (3.9.0)
Requirement already satisfied: typing-extensions>=4.10.0 in
c:\users\admin\appdata\roaming\python\python311\site-packages (from
torch>=1.8.0->ultralitics) (4.14.1)
Requirement already satisfied: sympy>=1.13.3 in
c:\users\admin\appdata\roaming\python\python311\site-packages (from
torch>=1.8.0->ultralitics) (1.14.0)
Requirement already satisfied: networkx in c:\programdata\anaconda3\lib\site-
packages (from torch>=1.8.0->ultralitics) (3.1)
Requirement already satisfied: jinja2 in c:\programdata\anaconda3\lib\site-
packages (from torch>=1.8.0->ultralitics) (3.1.2)
Requirement already satisfied: fsspec in c:\programdata\anaconda3\lib\site-
packages (from torch>=1.8.0->ultralitics) (2023.3.0)
Requirement already satisfied: six>=1.5 in c:\programdata\anaconda3\lib\site-
packages (from python-dateutil>=2.7->matplotlib>=3.3.0->ultralitics) (1.16.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
c:\programdata\anaconda3\lib\site-packages (from
sympy>=1.13.3->torch>=1.8.0->ultralitics) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
c:\programdata\anaconda3\lib\site-packages (from
jinja2->torch>=1.8.0->ultralitics) (2.1.1)
Downloading ultralitics-8.3.197-py3-none-any.whl (1.1 MB)
----- 0.0/1.1 MB ? eta -:-:--
----- 0.1/1.1 MB 4.2 MB/s eta 0:00:01
----- 0.3/1.1 MB 4.1 MB/s eta 0:00:01
----- 0.4/1.1 MB 3.1 MB/s eta 0:00:01
----- 0.4/1.1 MB 2.4 MB/s eta 0:00:01
----- 0.4/1.1 MB 2.4 MB/s eta 0:00:01
----- 0.5/1.1 MB 1.9 MB/s eta 0:00:01
----- 0.5/1.1 MB 1.6 MB/s eta 0:00:01
----- 0.6/1.1 MB 1.5 MB/s eta 0:00:01
----- 0.6/1.1 MB 1.5 MB/s eta 0:00:01
----- 0.6/1.1 MB 1.5 MB/s eta 0:00:01
----- 0.6/1.1 MB 1.5 MB/s eta 0:00:01

```

```

----- 0.6/1.1 MB 1.5 MB/s eta 0:00:01
----- 0.6/1.1 MB 1.5 MB/s eta 0:00:01
----- 0.6/1.1 MB 1.5 MB/s eta 0:00:01
----- 0.7/1.1 MB 1.1 MB/s eta 0:00:01
----- 0.7/1.1 MB 1.1 MB/s eta 0:00:01
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```

Installing collected packages: polars, ultralytics-thop, ultralytics
 Successfully installed polars-1.33.1 ultralytics-8.3.197 ultralytics-thop-2.0.17

DEPRECATION: Loading egg at c:\programdata\anaconda3\lib\site-packages\vboxapi-1.0-py3.11.egg is deprecated. pip 23.3 will enforce this behaviour change. A possible replacement is to use pip for package installation..

WARNING: The scripts ultralytics.exe and yolo.exe are installed in 'C:\Users\admin\AppData\Roaming\Python\Python311\Scripts' which is not on PATH. Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

```
[53]: from ultralytics import YOLO
import glob
import os
```

Creating new Ultralytics Settings v0.0.6 file
 View Ultralytics Settings with 'yolo settings' or at 'C:\Users\admin\AppData\Roaming\Ultralytics\settings.json'
 Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see <https://docs.ultralytics.com/quickstart/#ultralytics-settings>.

```
[54]: model = YOLO('yolov8n.pt')
```

Downloading
<https://github.com/ultralytics/assets/releases/download/v8.3.0/yolov8n.pt> to 'yolov8n.pt': 100% 6.2MB 592.7KB/s 10.8s.8s<0.2s6.1s

```
[55]: image_folder = 'E:/126156040/Object detection dataset/train/train'
output_folder = 'output1'
os.makedirs(output_folder, exist_ok=True)
```



```
[57]: # Get all PNG and JPG images
image_paths = glob.glob(os.path.join(image_folder, '*.png')) + glob.glob(os.
↳ path.join(image_folder, '*.jpg'))
```

```
[58]: # Loop through all images
for img_path in image_paths:
    r = model(img_path)
    # Save output image
    out_path = os.path.join(output_folder, os.path.basename(img_path))
    r[0].save(out_path)

    """print(f"Image: {img_path}")
    print("Boxes:", r[0].boxes.xyxy)
    print("Confidences:", r[0].boxes.conf)
    print("Classes:", r[0].boxes.cls)
    print("-" * 40)"""
```

```
image 1/1 E:\126156040\Object detection dataset\train\train\apple_1.jpg: 640x640
1 apple, 139.8ms
Speed: 11.7ms preprocess, 139.8ms inference, 17.2ms postprocess per image at
shape (1, 3, 640, 640)
```

```
image 1/1 E:\126156040\Object detection dataset\train\train\apple_10.jpg:
640x640 1 apple, 65.9ms
Speed: 4.0ms preprocess, 65.9ms inference, 0.8ms postprocess per image at shape
(1, 3, 640, 640)
```

```
image 1/1 E:\126156040\Object detection dataset\train\train\apple_11.jpg:
448x640 3 apples, 1 carrot, 64.6ms
Speed: 3.1ms preprocess, 64.6ms inference, 2.1ms postprocess per image at shape
(1, 3, 448, 640)
```

```
image 1/1 E:\126156040\Object detection dataset\train\train\apple_12.jpg:
640x640 2 apples, 73.1ms
Speed: 4.6ms preprocess, 73.1ms inference, 1.3ms postprocess per image at shape
(1, 3, 640, 640)
```

```
image 1/1 E:\126156040\Object detection dataset\train\train\apple_13.jpg:
640x640 3 apples, 71.9ms
Speed: 3.5ms preprocess, 71.9ms inference, 1.5ms postprocess per image at shape
(1, 3, 640, 640)
```

```
image 1/1 E:\126156040\Object detection dataset\train\train\apple_14.jpg:
448x640 2 apples, 1 orange, 54.1ms
Speed: 3.1ms preprocess, 54.1ms inference, 1.0ms postprocess per image at shape
(1, 3, 448, 640)
```

image 1/1 E:\126156040\Object detection dataset\train\train\apple_15.jpg:
480x640 1 apple, 69.9ms
Speed: 4.3ms preprocess, 69.9ms inference, 1.0ms postprocess per image at shape
(1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_16.jpg:
608x640 1 apple, 89.4ms
Speed: 3.3ms preprocess, 89.4ms inference, 1.1ms postprocess per image at shape
(1, 3, 608, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_17.jpg:
640x576 1 apple, 75.0ms
Speed: 4.8ms preprocess, 75.0ms inference, 1.1ms postprocess per image at shape
(1, 3, 640, 576)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_18.jpg:
512x640 (no detections), 71.0ms
Speed: 3.8ms preprocess, 71.0ms inference, 0.5ms postprocess per image at shape
(1, 3, 512, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_19.jpg:
480x640 1 apple, 58.6ms
Speed: 4.0ms preprocess, 58.6ms inference, 1.0ms postprocess per image at shape
(1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_2.jpg: 640x640
1 apple, 79.3ms
Speed: 2.9ms preprocess, 79.3ms inference, 1.0ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_20.jpg:
640x640 1 stop sign, 1 umbrella, 69.1ms
Speed: 3.2ms preprocess, 69.1ms inference, 1.0ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_21.jpg:
448x640 2 apples, 50.0ms
Speed: 1.0ms preprocess, 50.0ms inference, 0.9ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_22.jpg:
640x640 1 apple, 66.1ms
Speed: 4.5ms preprocess, 66.1ms inference, 0.7ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_23.jpg:
640x640 3 apples, 68.1ms

Speed: 3.1ms preprocess, 68.1ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_24.jpg:
640x448 1 bowl, 4 apples, 1 dining table, 73.2ms
Speed: 3.5ms preprocess, 73.2ms inference, 2.2ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_25.jpg:
448x640 2 apples, 59.7ms
Speed: 2.5ms preprocess, 59.7ms inference, 1.4ms postprocess per image at shape (1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_26.jpg:
448x640 2 apples, 52.9ms
Speed: 2.1ms preprocess, 52.9ms inference, 1.2ms postprocess per image at shape (1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_27.jpg:
320x640 3 apples, 51.0ms
Speed: 1.2ms preprocess, 51.0ms inference, 1.3ms postprocess per image at shape (1, 3, 320, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_28.jpg:
640x640 1 apple, 68.0ms
Speed: 4.4ms preprocess, 68.0ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_29.jpg:
640x640 1 apple, 69.3ms
Speed: 3.4ms preprocess, 69.3ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_3.jpg: 480x640
17 apples, 57.1ms
Speed: 2.6ms preprocess, 57.1ms inference, 4.1ms postprocess per image at shape (1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_30.jpg:
640x640 4 apples, 65.1ms
Speed: 4.2ms preprocess, 65.1ms inference, 1.2ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_31.jpg:
384x640 3 apples, 1 orange, 59.2ms
Speed: 2.1ms preprocess, 59.2ms inference, 1.4ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_32.jpg:
480x640 2 apples, 60.6ms
Speed: 2.6ms preprocess, 60.6ms inference, 1.3ms postprocess per image at shape
(1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_33.jpg:
640x640 1 apple, 79.3ms
Speed: 5.6ms preprocess, 79.3ms inference, 1.1ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_35.jpg:
640x448 1 apple, 58.0ms
Speed: 2.5ms preprocess, 58.0ms inference, 1.2ms postprocess per image at shape
(1, 3, 640, 448)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_36.jpg:
640x640 (no detections), 65.3ms
Speed: 3.5ms preprocess, 65.3ms inference, 0.4ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_37.jpg:
608x640 1 apple, 71.3ms
Speed: 3.1ms preprocess, 71.3ms inference, 1.3ms postprocess per image at shape
(1, 3, 608, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_38.jpg:
640x608 1 kite, 1 apple, 2 oranges, 89.0ms
Speed: 5.1ms preprocess, 89.0ms inference, 1.3ms postprocess per image at shape
(1, 3, 640, 608)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_39.jpg:
384x640 1 apple, 47.5ms
Speed: 2.1ms preprocess, 47.5ms inference, 1.0ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_4.jpg: 448x640
2 apples, 49.8ms
Speed: 1.5ms preprocess, 49.8ms inference, 1.1ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_40.jpg:
416x640 2 apples, 2 oranges, 63.2ms
Speed: 2.6ms preprocess, 63.2ms inference, 1.5ms postprocess per image at shape
(1, 3, 416, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_41.jpg:
544x640 1 vase, 79.8ms
Speed: 3.4ms preprocess, 79.8ms inference, 1.0ms postprocess per image at shape

(1, 3, 544, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_42.jpg:
640x640 1 apple, 77.5ms
Speed: 4.6ms preprocess, 77.5ms inference, 0.8ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_43.jpg:
640x640 (no detections), 70.3ms
Speed: 5.2ms preprocess, 70.3ms inference, 0.6ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_44.jpg:
640x640 1 apple, 79.4ms
Speed: 4.1ms preprocess, 79.4ms inference, 1.0ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_45.jpg:
512x640 2 apples, 1 orange, 66.6ms
Speed: 3.9ms preprocess, 66.6ms inference, 1.4ms postprocess per image at shape
(1, 3, 512, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_46.jpg:
480x640 6 apples, 64.4ms
Speed: 2.7ms preprocess, 64.4ms inference, 1.4ms postprocess per image at shape
(1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_47.jpg:
448x640 4 apples, 52.6ms
Speed: 2.5ms preprocess, 52.6ms inference, 1.5ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_48.jpg:
448x640 2 apples, 50.1ms
Speed: 1.7ms preprocess, 50.1ms inference, 0.9ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_49.jpg:
640x608 1 cake, 64.4ms
Speed: 2.6ms preprocess, 64.4ms inference, 0.8ms postprocess per image at shape
(1, 3, 640, 608)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_5.jpg: 384x640
8 apples, 49.3ms
Speed: 1.8ms preprocess, 49.3ms inference, 2.3ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_50.jpg:

512x640 4 apples, 61.7ms
Speed: 2.0ms preprocess, 61.7ms inference, 1.8ms postprocess per image at shape (1, 3, 512, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_51.jpg:
640x576 1 apple, 75.7ms
Speed: 3.3ms preprocess, 75.7ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 576)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_52.jpg:
640x640 6 apples, 70.8ms
Speed: 4.6ms preprocess, 70.8ms inference, 1.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_53.jpg:
576x640 1 apple, 77.1ms
Speed: 3.2ms preprocess, 77.1ms inference, 1.1ms postprocess per image at shape (1, 3, 576, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_54.jpg:
608x640 1 apple, 1 donut, 77.0ms
Speed: 3.6ms preprocess, 77.0ms inference, 1.3ms postprocess per image at shape (1, 3, 608, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_55.jpg:
448x640 6 apples, 2 oranges, 60.4ms
Speed: 3.0ms preprocess, 60.4ms inference, 2.3ms postprocess per image at shape (1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_56.jpg:
480x640 1 apple, 63.6ms
Speed: 3.3ms preprocess, 63.6ms inference, 1.0ms postprocess per image at shape (1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_57.jpg:
448x640 1 apple, 53.8ms
Speed: 1.9ms preprocess, 53.8ms inference, 1.0ms postprocess per image at shape (1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_58.jpg:
640x576 4 apples, 63.4ms
Speed: 3.4ms preprocess, 63.4ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 576)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_59.jpg:
480x640 2 apples, 1 orange, 52.7ms
Speed: 2.2ms preprocess, 52.7ms inference, 1.2ms postprocess per image at shape (1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_6.jpg: 416x640
3 apples, 1 orange, 50.4ms
Speed: 2.2ms preprocess, 50.4ms inference, 1.4ms postprocess per image at shape
(1, 3, 416, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_60.jpg:
640x640 8 apples, 69.2ms
Speed: 3.7ms preprocess, 69.2ms inference, 2.3ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_61.jpg:
640x640 2 apples, 1 orange, 66.9ms
Speed: 5.1ms preprocess, 66.9ms inference, 1.1ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_62.jpg:
448x640 1 apple, 52.5ms
Speed: 2.5ms preprocess, 52.5ms inference, 0.8ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_63.jpg:
384x640 4 apples, 1 orange, 54.2ms
Speed: 2.1ms preprocess, 54.2ms inference, 1.3ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_64.jpg:
640x640 3 apples, 67.3ms
Speed: 3.0ms preprocess, 67.3ms inference, 1.1ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_65.jpg:
480x640 5 apples, 57.1ms
Speed: 1.9ms preprocess, 57.1ms inference, 1.4ms postprocess per image at shape
(1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_66.jpg:
480x640 3 apples, 64.1ms
Speed: 2.7ms preprocess, 64.1ms inference, 1.6ms postprocess per image at shape
(1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_67.jpg:
512x640 3 apples, 64.0ms
Speed: 2.4ms preprocess, 64.0ms inference, 1.4ms postprocess per image at shape
(1, 3, 512, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_68.jpg:
640x640 2 apples, 65.0ms

Speed: 3.7ms preprocess, 65.0ms inference, 1.2ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_69.jpg:

640x640 3 apples, 73.8ms

Speed: 4.8ms preprocess, 73.8ms inference, 1.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_7.jpg: 640x640
1 apple, 74.6ms

Speed: 6.1ms preprocess, 74.6ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_70.jpg:

640x512 1 apple, 73.8ms

Speed: 2.5ms preprocess, 73.8ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 512)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_71.jpg:

384x640 4 apples, 50.4ms

Speed: 2.5ms preprocess, 50.4ms inference, 1.6ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_72.jpg:

640x640 1 apple, 75.5ms

Speed: 3.0ms preprocess, 75.5ms inference, 1.3ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_73.jpg:

640x640 1 apple, 68.4ms

Speed: 3.9ms preprocess, 68.4ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_74.jpg:

640x640 2 apples, 66.5ms

Speed: 3.4ms preprocess, 66.5ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_75.jpg:

640x640 1 apple, 67.9ms

Speed: 3.4ms preprocess, 67.9ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_76.jpg:

512x640 2 apples, 72.0ms

Speed: 13.0ms preprocess, 72.0ms inference, 1.4ms postprocess per image at shape (1, 3, 512, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_8.jpg: 480x640
1 apple, 61.4ms
Speed: 2.0ms preprocess, 61.4ms inference, 1.1ms postprocess per image at shape
(1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\apple_9.jpg: 480x640
1 apple, 52.9ms
Speed: 2.1ms preprocess, 52.9ms inference, 0.8ms postprocess per image at shape
(1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_1.jpg:
384x640 1 banana, 47.2ms
Speed: 1.9ms preprocess, 47.2ms inference, 0.9ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_10.jpg:
448x640 4 bananas, 51.6ms
Speed: 2.2ms preprocess, 51.6ms inference, 1.3ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_11.jpg:
640x640 1 scissors, 64.4ms
Speed: 3.5ms preprocess, 64.4ms inference, 0.7ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_12.jpg:
640x640 1 banana, 72.0ms
Speed: 2.4ms preprocess, 72.0ms inference, 1.0ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_13.jpg:
640x640 1 kite, 85.4ms
Speed: 8.0ms preprocess, 85.4ms inference, 1.6ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_14.jpg:
512x640 1 bowl, 1 banana, 1 dining table, 57.8ms
Speed: 2.4ms preprocess, 57.8ms inference, 0.9ms postprocess per image at shape
(1, 3, 512, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_16.jpg:
384x640 1 banana, 1 scissors, 46.1ms
Speed: 2.1ms preprocess, 46.1ms inference, 0.9ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_17.jpg:
640x640 4 bananas, 71.1ms
Speed: 2.4ms preprocess, 71.1ms inference, 1.4ms postprocess per image at shape

(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_2.jpg:
544x640 (no detections), 60.5ms
Speed: 4.7ms preprocess, 60.5ms inference, 0.5ms postprocess per image at shape
(1, 3, 544, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_20.jpg:
384x640 1 banana, 52.2ms
Speed: 1.3ms preprocess, 52.2ms inference, 1.2ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_21.jpg:
384x640 1 banana, 52.1ms
Speed: 1.9ms preprocess, 52.1ms inference, 0.9ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_22.jpg:
448x640 4 bananas, 58.6ms
Speed: 1.7ms preprocess, 58.6ms inference, 1.5ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_23.jpg:
352x640 1 banana, 59.3ms
Speed: 2.1ms preprocess, 59.3ms inference, 0.9ms postprocess per image at shape
(1, 3, 352, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_24.jpg:
352x640 1 banana, 47.5ms
Speed: 0.9ms preprocess, 47.5ms inference, 0.7ms postprocess per image at shape
(1, 3, 352, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_25.jpg:
640x448 2 bananas, 64.1ms
Speed: 2.2ms preprocess, 64.1ms inference, 1.1ms postprocess per image at shape
(1, 3, 640, 448)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_26.jpg:
352x640 1 banana, 46.5ms
Speed: 1.4ms preprocess, 46.5ms inference, 0.9ms postprocess per image at shape
(1, 3, 352, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_27.jpg:
512x640 4 bananas, 54.3ms
Speed: 3.5ms preprocess, 54.3ms inference, 1.4ms postprocess per image at shape
(1, 3, 512, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_28.jpg:

640x640 1 banana, 66.3ms
Speed: 5.0ms preprocess, 66.3ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_29.jpg:
480x640 1 banana, 56.0ms
Speed: 2.6ms preprocess, 56.0ms inference, 1.0ms postprocess per image at shape (1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_3.jpg:
448x640 1 banana, 54.8ms
Speed: 2.4ms preprocess, 54.8ms inference, 0.7ms postprocess per image at shape (1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_30.jpg:
512x640 2 bananas, 56.1ms
Speed: 1.9ms preprocess, 56.1ms inference, 1.0ms postprocess per image at shape (1, 3, 512, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_31.jpg:
384x640 1 banana, 47.9ms
Speed: 1.4ms preprocess, 47.9ms inference, 0.8ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_32.jpg:
544x640 3 bananas, 66.7ms
Speed: 2.2ms preprocess, 66.7ms inference, 1.4ms postprocess per image at shape (1, 3, 544, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_33.jpg:
480x640 (no detections), 60.3ms
Speed: 2.9ms preprocess, 60.3ms inference, 0.4ms postprocess per image at shape (1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_34.jpg:
640x608 2 bananas, 84.7ms
Speed: 3.5ms preprocess, 84.7ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 608)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_35.jpg:
448x640 1 banana, 2 apples, 2 oranges, 53.7ms
Speed: 1.4ms preprocess, 53.7ms inference, 1.5ms postprocess per image at shape (1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_36.jpg:
416x640 1 banana, 47.2ms
Speed: 2.1ms preprocess, 47.2ms inference, 0.7ms postprocess per image at shape (1, 3, 416, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_37.jpg:
384x640 9 bananas, 51.6ms
Speed: 1.4ms preprocess, 51.6ms inference, 2.1ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_38.jpg:
544x640 2 bananas, 67.6ms
Speed: 3.4ms preprocess, 67.6ms inference, 1.2ms postprocess per image at shape
(1, 3, 544, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_39.jpg:
640x640 1 banana, 75.3ms
Speed: 4.3ms preprocess, 75.3ms inference, 1.2ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_4.jpg:
448x640 2 bananas, 1 dining table, 57.2ms
Speed: 2.7ms preprocess, 57.2ms inference, 1.2ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_40.jpg:
448x640 1 banana, 50.9ms
Speed: 1.7ms preprocess, 50.9ms inference, 1.1ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_41.jpg:
384x640 1 banana, 52.6ms
Speed: 2.5ms preprocess, 52.6ms inference, 1.2ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_42.jpg:
352x640 1 scissors, 50.7ms
Speed: 2.1ms preprocess, 50.7ms inference, 0.9ms postprocess per image at shape
(1, 3, 352, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_43.jpg:
448x640 4 bananas, 51.4ms
Speed: 2.0ms preprocess, 51.4ms inference, 1.1ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_44.jpg:
448x640 1 banana, 57.4ms
Speed: 1.5ms preprocess, 57.4ms inference, 1.1ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_45.jpg:
640x608 2 bananas, 75.2ms

Speed: 4.2ms preprocess, 75.2ms inference, 1.3ms postprocess per image at shape (1, 3, 640, 608)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_46.jpg:
320x640 1 banana, 49.5ms
Speed: 1.1ms preprocess, 49.5ms inference, 1.0ms postprocess per image at shape (1, 3, 320, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_47.jpg:
512x640 (no detections), 63.3ms
Speed: 2.3ms preprocess, 63.3ms inference, 0.8ms postprocess per image at shape (1, 3, 512, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_48.jpg:
384x640 6 bananas, 1 dining table, 48.4ms
Speed: 1.9ms preprocess, 48.4ms inference, 2.0ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_49.jpg:
320x640 1 banana, 46.2ms
Speed: 1.6ms preprocess, 46.2ms inference, 0.9ms postprocess per image at shape (1, 3, 320, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_5.jpg:
640x448 3 bananas, 55.6ms
Speed: 2.0ms preprocess, 55.6ms inference, 1.2ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_50.jpg:
448x640 5 bananas, 57.6ms
Speed: 2.3ms preprocess, 57.6ms inference, 1.8ms postprocess per image at shape (1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_51.jpg:
352x640 1 banana, 47.8ms
Speed: 1.9ms preprocess, 47.8ms inference, 0.9ms postprocess per image at shape (1, 3, 352, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_52.jpg:
256x640 4 bananas, 49.8ms
Speed: 1.0ms preprocess, 49.8ms inference, 1.5ms postprocess per image at shape (1, 3, 256, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_53.jpg:
640x608 (no detections), 91.3ms
Speed: 3.6ms preprocess, 91.3ms inference, 0.6ms postprocess per image at shape (1, 3, 640, 608)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_54.jpg:
480x640 1 banana, 54.4ms
Speed: 2.6ms preprocess, 54.4ms inference, 1.5ms postprocess per image at shape
(1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_55.jpg:
640x416 2 bananas, 62.9ms
Speed: 2.4ms preprocess, 62.9ms inference, 0.7ms postprocess per image at shape
(1, 3, 640, 416)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_56.jpg:
352x640 1 banana, 48.5ms
Speed: 1.6ms preprocess, 48.5ms inference, 1.0ms postprocess per image at shape
(1, 3, 352, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_57.jpg:
640x384 (no detections), 62.7ms
Speed: 2.7ms preprocess, 62.7ms inference, 0.5ms postprocess per image at shape
(1, 3, 640, 384)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_58.jpg:
320x640 6 bananas, 1 apple, 1 orange, 1 cake, 43.4ms
Speed: 2.1ms preprocess, 43.4ms inference, 2.2ms postprocess per image at shape
(1, 3, 320, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_59.jpg:
448x640 3 persons, 1 banana, 59.5ms
Speed: 2.8ms preprocess, 59.5ms inference, 1.4ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_6.jpg:
448x640 1 person, 2 bananas, 47.9ms
Speed: 2.1ms preprocess, 47.9ms inference, 0.9ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_60.jpg:
512x640 1 banana, 62.9ms
Speed: 2.1ms preprocess, 62.9ms inference, 1.0ms postprocess per image at shape
(1, 3, 512, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_61.jpg:
384x640 1 banana, 1 apple, 44.5ms
Speed: 0.9ms preprocess, 44.5ms inference, 0.9ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_62.jpg:
384x640 1 banana, 46.8ms
Speed: 1.4ms preprocess, 46.8ms inference, 1.0ms postprocess per image at shape
(1, 3, 384, 640)

(1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_63.jpg:
576x640 5 bananas, 70.9ms
Speed: 3.3ms preprocess, 70.9ms inference, 1.4ms postprocess per image at shape
(1, 3, 576, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_64.jpg:
416x640 1 banana, 59.5ms
Speed: 1.5ms preprocess, 59.5ms inference, 1.0ms postprocess per image at shape
(1, 3, 416, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_65.jpg:
448x640 1 bowl, 7 bananas, 52.1ms
Speed: 1.3ms preprocess, 52.1ms inference, 2.3ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_66.jpg:
448x640 1 banana, 46.9ms
Speed: 1.5ms preprocess, 46.9ms inference, 0.7ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_67.jpg:
640x640 1 toilet, 75.5ms
Speed: 3.9ms preprocess, 75.5ms inference, 0.7ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_68.jpg:
384x640 7 bananas, 44.9ms
Speed: 2.2ms preprocess, 44.9ms inference, 1.4ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_69.jpg:
320x640 1 banana, 40.1ms
Speed: 1.6ms preprocess, 40.1ms inference, 0.7ms postprocess per image at shape
(1, 3, 320, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_7.jpg:
640x480 2 bananas, 1 teddy bear, 64.6ms
Speed: 1.8ms preprocess, 64.6ms inference, 1.5ms postprocess per image at shape
(1, 3, 640, 480)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_70.jpg:
320x640 1 banana, 43.2ms
Speed: 1.4ms preprocess, 43.2ms inference, 1.0ms postprocess per image at shape
(1, 3, 320, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_71.jpg:

320x640 1 banana, 41.4ms
Speed: 1.3ms preprocess, 41.4ms inference, 1.0ms postprocess per image at shape (1, 3, 320, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_72.jpg:
448x640 1 frisbee, 51.4ms
Speed: 2.3ms preprocess, 51.4ms inference, 0.7ms postprocess per image at shape (1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_73.jpg:
384x640 1 banana, 55.6ms
Speed: 2.9ms preprocess, 55.6ms inference, 1.2ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_74.jpg:
448x640 4 bananas, 54.8ms
Speed: 2.6ms preprocess, 54.8ms inference, 1.3ms postprocess per image at shape (1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_75.jpg:
416x640 1 kite, 49.5ms
Speed: 1.7ms preprocess, 49.5ms inference, 0.9ms postprocess per image at shape (1, 3, 416, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_76.jpg:
384x640 1 banana, 47.9ms
Speed: 1.7ms preprocess, 47.9ms inference, 1.0ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_8.jpg:
448x640 4 bananas, 52.9ms
Speed: 2.6ms preprocess, 52.9ms inference, 1.1ms postprocess per image at shape (1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\banana_9.jpg:
640x480 1 person, 1 banana, 55.9ms
Speed: 3.0ms preprocess, 55.9ms inference, 1.2ms postprocess per image at shape (1, 3, 640, 480)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_1.jpg: 480x640
1 banana, 1 apple, 1 orange, 62.1ms
Speed: 3.0ms preprocess, 62.1ms inference, 1.4ms postprocess per image at shape (1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_10.jpg:
448x640 1 banana, 1 apple, 1 orange, 52.9ms
Speed: 1.7ms preprocess, 52.9ms inference, 1.5ms postprocess per image at shape (1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_11.jpg:
480x640 2 bananas, 2 apples, 1 orange, 1 dining table, 53.7ms
Speed: 2.0ms preprocess, 53.7ms inference, 1.3ms postprocess per image at shape
(1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_12.jpg:
416x640 1 banana, 1 apple, 1 orange, 48.0ms
Speed: 2.3ms preprocess, 48.0ms inference, 1.0ms postprocess per image at shape
(1, 3, 416, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_13.jpg:
480x640 2 bananas, 2 apples, 1 orange, 51.4ms
Speed: 2.8ms preprocess, 51.4ms inference, 1.3ms postprocess per image at shape
(1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_14.jpg:
576x640 1 banana, 1 apple, 58.2ms
Speed: 3.3ms preprocess, 58.2ms inference, 1.1ms postprocess per image at shape
(1, 3, 576, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_15.jpg:
448x640 1 banana, 1 apple, 1 orange, 49.7ms
Speed: 2.0ms preprocess, 49.7ms inference, 1.0ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_16.jpg:
608x640 1 apple, 4 oranges, 79.1ms
Speed: 2.9ms preprocess, 79.1ms inference, 1.7ms postprocess per image at shape
(1, 3, 608, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_17.jpg:
448x640 3 bananas, 9 apples, 5 oranges, 47.5ms
Speed: 2.3ms preprocess, 47.5ms inference, 2.8ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_18.jpg:
448x640 2 bananas, 2 oranges, 49.1ms
Speed: 1.8ms preprocess, 49.1ms inference, 1.6ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_19.jpg:
608x640 1 bowl, 1 banana, 1 apple, 7 oranges, 71.9ms
Speed: 3.8ms preprocess, 71.9ms inference, 2.6ms postprocess per image at shape
(1, 3, 608, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_2.jpg: 480x640
1 banana, 2 apples, 1 orange, 59.7ms

Speed: 2.5ms preprocess, 59.7ms inference, 1.6ms postprocess per image at shape (1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_20.jpg:

576x640 1 orange, 67.9ms

Speed: 3.4ms preprocess, 67.9ms inference, 1.3ms postprocess per image at shape (1, 3, 576, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_3.jpg: 544x640

1 banana, 2 apples, 1 orange, 71.5ms

Speed: 3.4ms preprocess, 71.5ms inference, 1.6ms postprocess per image at shape (1, 3, 544, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_4.jpg: 640x480

1 banana, 1 apple, 1 orange, 54.7ms

Speed: 2.7ms preprocess, 54.7ms inference, 1.5ms postprocess per image at shape (1, 3, 640, 480)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_5.jpg: 640x640

1 banana, 1 apple, 1 orange, 68.1ms

Speed: 3.5ms preprocess, 68.1ms inference, 1.4ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_6.jpg: 448x640

1 banana, 1 apple, 2 oranges, 49.9ms

Speed: 1.9ms preprocess, 49.9ms inference, 1.1ms postprocess per image at shape (1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_7.jpg: 416x640

1 banana, 2 apples, 1 orange, 50.3ms

Speed: 1.6ms preprocess, 50.3ms inference, 1.5ms postprocess per image at shape (1, 3, 416, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_8.jpg: 608x640

1 banana, 1 apple, 1 orange, 65.3ms

Speed: 3.5ms preprocess, 65.3ms inference, 1.6ms postprocess per image at shape (1, 3, 608, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\mixed_9.jpg: 640x640

3 umbrellas, 1 kite, 71.1ms

Speed: 3.8ms preprocess, 71.1ms inference, 1.6ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_1.jpg:

352x640 10 oranges, 46.8ms

Speed: 1.0ms preprocess, 46.8ms inference, 2.0ms postprocess per image at shape (1, 3, 352, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_10.jpg:
384x640 3 oranges, 51.0ms
Speed: 1.4ms preprocess, 51.0ms inference, 1.0ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_11.jpg:
640x640 1 bowl, 1 cake, 79.8ms
Speed: 4.7ms preprocess, 79.8ms inference, 0.9ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_12.jpg:
352x640 1 umbrella, 1 apple, 1 orange, 47.2ms
Speed: 1.1ms preprocess, 47.2ms inference, 1.2ms postprocess per image at shape
(1, 3, 352, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_13.jpg:
608x640 1 apple, 1 orange, 74.9ms
Speed: 2.7ms preprocess, 74.9ms inference, 1.3ms postprocess per image at shape
(1, 3, 608, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_14.jpg:
640x640 1 orange, 71.8ms
Speed: 3.5ms preprocess, 71.8ms inference, 1.0ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_15.jpg:
448x640 13 oranges, 54.1ms
Speed: 2.2ms preprocess, 54.1ms inference, 3.0ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_16.jpg:
640x640 1 orange, 64.5ms
Speed: 3.1ms preprocess, 64.5ms inference, 0.8ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_17.jpg:
448x640 5 oranges, 53.6ms
Speed: 2.7ms preprocess, 53.6ms inference, 2.0ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_18.jpg:
640x640 2 oranges, 75.0ms
Speed: 3.0ms preprocess, 75.0ms inference, 1.4ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_19.jpg:
480x640 11 oranges, 54.0ms
Speed: 2.2ms preprocess, 54.0ms inference, 2.1ms postprocess per image at shape

(1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_2.jpg:
576x640 3 oranges, 69.5ms
Speed: 3.9ms preprocess, 69.5ms inference, 1.5ms postprocess per image at shape
(1, 3, 576, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_20.jpg:
608x640 1 orange, 80.5ms
Speed: 5.2ms preprocess, 80.5ms inference, 0.9ms postprocess per image at shape
(1, 3, 608, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_21.jpg:
384x640 1 bird, 5 oranges, 45.7ms
Speed: 2.6ms preprocess, 45.7ms inference, 2.0ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_22.jpg:
384x640 1 orange, 48.4ms
Speed: 2.0ms preprocess, 48.4ms inference, 1.2ms postprocess per image at shape
(1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_23.jpg:
640x640 6 oranges, 71.2ms
Speed: 3.2ms preprocess, 71.2ms inference, 1.9ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_24.jpg:
448x640 6 oranges, 1 dining table, 51.0ms
Speed: 2.7ms preprocess, 51.0ms inference, 1.9ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_25.jpg:
640x640 1 orange, 67.8ms
Speed: 3.4ms preprocess, 67.8ms inference, 0.8ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_26.jpg:
640x640 4 oranges, 75.8ms
Speed: 12.9ms preprocess, 75.8ms inference, 1.4ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_27.jpg:
448x640 4 oranges, 51.5ms
Speed: 1.5ms preprocess, 51.5ms inference, 1.5ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_28.jpg:

640x640 1 orange, 68.7ms
Speed: 4.0ms preprocess, 68.7ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_29.jpg:
448x640 4 oranges, 1 dining table, 54.9ms
Speed: 1.8ms preprocess, 54.9ms inference, 1.6ms postprocess per image at shape (1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_3.jpg:
640x640 10 oranges, 79.2ms
Speed: 4.0ms preprocess, 79.2ms inference, 3.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_30.jpg:
640x640 1 orange, 76.1ms
Speed: 3.6ms preprocess, 76.1ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_31.jpg:
608x640 4 oranges, 63.7ms
Speed: 3.4ms preprocess, 63.7ms inference, 1.3ms postprocess per image at shape (1, 3, 608, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_32.jpg:
480x640 3 oranges, 56.0ms
Speed: 1.7ms preprocess, 56.0ms inference, 1.4ms postprocess per image at shape (1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_33.jpg:
384x640 1 umbrella, 1 orange, 48.7ms
Speed: 1.5ms preprocess, 48.7ms inference, 1.1ms postprocess per image at shape (1, 3, 384, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_34.jpg:
480x640 2 oranges, 55.9ms
Speed: 3.1ms preprocess, 55.9ms inference, 1.3ms postprocess per image at shape (1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_35.jpg:
640x640 (no detections), 74.0ms
Speed: 3.2ms preprocess, 74.0ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_36.jpg:
480x640 6 oranges, 56.1ms
Speed: 3.1ms preprocess, 56.1ms inference, 1.3ms postprocess per image at shape (1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_37.jpg:
544x640 1 apple, 1 orange, 62.3ms
Speed: 2.5ms preprocess, 62.3ms inference, 0.9ms postprocess per image at shape
(1, 3, 544, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_38.jpg:
640x640 5 oranges, 70.9ms
Speed: 3.1ms preprocess, 70.9ms inference, 1.6ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_4.jpg:
640x640 1 orange, 73.1ms
Speed: 5.0ms preprocess, 73.1ms inference, 0.8ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_40.jpg:
480x640 3 oranges, 57.7ms
Speed: 3.3ms preprocess, 57.7ms inference, 1.4ms postprocess per image at shape
(1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_41.jpg:
416x640 4 oranges, 53.0ms
Speed: 2.0ms preprocess, 53.0ms inference, 1.4ms postprocess per image at shape
(1, 3, 416, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_42.jpg:
640x640 (no detections), 70.4ms
Speed: 5.5ms preprocess, 70.4ms inference, 0.5ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_43.jpg:
640x640 3 oranges, 70.3ms
Speed: 2.9ms preprocess, 70.3ms inference, 1.1ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_44.jpg:
480x640 1 orange, 58.1ms
Speed: 2.1ms preprocess, 58.1ms inference, 1.0ms postprocess per image at shape
(1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_46.jpg:
608x640 1 orange, 68.2ms
Speed: 2.9ms preprocess, 68.2ms inference, 1.1ms postprocess per image at shape
(1, 3, 608, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_47.jpg:
544x640 2 oranges, 63.0ms

Speed: 3.0ms preprocess, 63.0ms inference, 1.2ms postprocess per image at shape (1, 3, 544, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_48.jpg:
480x640 3 oranges, 59.4ms
Speed: 1.7ms preprocess, 59.4ms inference, 1.4ms postprocess per image at shape (1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_49.jpg:
640x608 1 umbrella, 1 kite, 2 oranges, 95.7ms
Speed: 4.7ms preprocess, 95.7ms inference, 1.8ms postprocess per image at shape (1, 3, 640, 608)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_5.jpg:
512x640 1 apple, 2 oranges, 58.4ms
Speed: 2.4ms preprocess, 58.4ms inference, 0.9ms postprocess per image at shape (1, 3, 512, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_50.jpg:
480x640 3 oranges, 56.8ms
Speed: 2.4ms preprocess, 56.8ms inference, 1.4ms postprocess per image at shape (1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_51.jpg:
448x640 3 oranges, 52.3ms
Speed: 1.6ms preprocess, 52.3ms inference, 0.9ms postprocess per image at shape (1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_52.jpg:
640x640 2 oranges, 73.2ms
Speed: 3.2ms preprocess, 73.2ms inference, 1.3ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_53.jpg:
608x640 1 orange, 73.7ms
Speed: 3.6ms preprocess, 73.7ms inference, 0.9ms postprocess per image at shape (1, 3, 608, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_54.jpg:
640x640 5 oranges, 77.0ms
Speed: 3.9ms preprocess, 77.0ms inference, 1.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_55.jpg:
480x640 1 orange, 61.7ms
Speed: 2.2ms preprocess, 61.7ms inference, 1.0ms postprocess per image at shape (1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_56.jpg:
544x640 5 oranges, 56.9ms
Speed: 4.5ms preprocess, 56.9ms inference, 1.2ms postprocess per image at shape
(1, 3, 544, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_57.jpg:
640x480 6 oranges, 58.1ms
Speed: 2.2ms preprocess, 58.1ms inference, 1.8ms postprocess per image at shape
(1, 3, 640, 480)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_58.jpg:
352x640 3 oranges, 54.6ms
Speed: 1.4ms preprocess, 54.6ms inference, 1.7ms postprocess per image at shape
(1, 3, 352, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_59.jpg:
640x640 3 oranges, 72.1ms
Speed: 3.3ms preprocess, 72.1ms inference, 1.4ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_6.jpg:
480x640 1 orange, 56.3ms
Speed: 2.2ms preprocess, 56.3ms inference, 1.0ms postprocess per image at shape
(1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_60.jpg:
640x608 (no detections), 61.8ms
Speed: 3.8ms preprocess, 61.8ms inference, 0.5ms postprocess per image at shape
(1, 3, 640, 608)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_61.jpg:
640x640 4 oranges, 72.3ms
Speed: 5.1ms preprocess, 72.3ms inference, 1.0ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_62.jpg:
352x640 5 oranges, 45.1ms
Speed: 1.8ms preprocess, 45.1ms inference, 1.5ms postprocess per image at shape
(1, 3, 352, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_63.jpg:
448x640 2 oranges, 49.6ms
Speed: 2.2ms preprocess, 49.6ms inference, 0.9ms postprocess per image at shape
(1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_64.jpg:
640x640 2 oranges, 76.0ms
Speed: 3.5ms preprocess, 76.0ms inference, 1.6ms postprocess per image at shape

(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_67.jpg:
352x640 2 oranges, 46.8ms
Speed: 1.5ms preprocess, 46.8ms inference, 0.7ms postprocess per image at shape
(1, 3, 352, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_68.jpg:
512x640 1 orange, 58.8ms
Speed: 2.8ms preprocess, 58.8ms inference, 0.7ms postprocess per image at shape
(1, 3, 512, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_69.jpg:
416x640 1 wine glass, 6 oranges, 48.5ms
Speed: 1.9ms preprocess, 48.5ms inference, 1.9ms postprocess per image at shape
(1, 3, 416, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_7.jpg:
352x640 2 oranges, 49.1ms
Speed: 2.3ms preprocess, 49.1ms inference, 0.9ms postprocess per image at shape
(1, 3, 352, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_70.jpg:
480x640 4 oranges, 60.5ms
Speed: 2.6ms preprocess, 60.5ms inference, 1.5ms postprocess per image at shape
(1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_71.jpg:
576x640 1 orange, 74.4ms
Speed: 3.6ms preprocess, 74.4ms inference, 1.1ms postprocess per image at shape
(1, 3, 576, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_72.jpg:
640x640 1 orange, 78.0ms
Speed: 3.2ms preprocess, 78.0ms inference, 1.0ms postprocess per image at shape
(1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_73.jpg:
480x640 1 orange, 57.4ms
Speed: 2.6ms preprocess, 57.4ms inference, 1.3ms postprocess per image at shape
(1, 3, 480, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_74.jpg:
512x640 4 oranges, 68.8ms
Speed: 2.4ms preprocess, 68.8ms inference, 1.5ms postprocess per image at shape
(1, 3, 512, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_75.jpg:

640x640 1 orange, 75.3ms

Speed: 4.4ms preprocess, 75.3ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_76.jpg:

448x640 10 oranges, 56.8ms

Speed: 4.6ms preprocess, 56.8ms inference, 2.7ms postprocess per image at shape (1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_8.jpg:

448x640 4 oranges, 58.6ms

Speed: 3.4ms preprocess, 58.6ms inference, 1.4ms postprocess per image at shape (1, 3, 448, 640)

image 1/1 E:\126156040\Object detection dataset\train\train\orange_9.jpg:

576x640 3 oranges, 60.6ms

Speed: 3.5ms preprocess, 60.6ms inference, 1.2ms postprocess per image at shape (1, 3, 576, 640)

```
[59]: import os
      from IPython.display import Image, display
```

```
[60]: # Get a list of files in the output folder
      output_files = os.listdir(output_folder)

      # Filter for image files (assuming they are .jpg or .png)
      image_files = [f for f in output_files if f.endswith('.jpg') or f.endswith('.
      ↪png')]
```

```
[61]: # Display the first image found, if any
      if image_files:
          first_image_path = os.path.join(output_folder, image_files[0])
          print(f"Displaying: {first_image_path}")
          display(Image(filename=first_image_path))
      else:
          print("No image files found in the output folder.")
```

Displaying: output1\apple_1.jpg



4 Exe 4(Banana dataset)

```
[41]: import tensorflow as tf
      from tensorflow.keras.preprocessing.image import ImageDataGenerator
      from tensorflow.keras.layers import Conv2D, MaxPooling2D, Conv2DTranspose, Input
      from tensorflow.keras.models import Model
      import matplotlib.pyplot as plt
      import numpy as np
```

```
# Paths to images and masks directories
image_dir = "E:/126156040/Banana FCN/Images"
mask_dir = "E:/126156040/Banana FCN/Mask"
```

```
[33]: # Image and mask data generators
      image_datagen = ImageDataGenerator(rescale=1./255)
```

```
mask_datagen = ImageDataGenerator(rescale=1./255)
```

```
[34]: image_generator = image_datagen.flow_from_directory(
    image_dir,
    class_mode=None,
    color_mode='rgb',
    target_size=(128, 128),
    batch_size=32,
    seed=42
)

mask_generator = mask_datagen.flow_from_directory(
    mask_dir,
    class_mode=None,
    color_mode='grayscale',
    target_size=(128, 128),
    batch_size=32,
    seed=42
)
```

Found 82 images belonging to 1 classes.

Found 82 images belonging to 1 classes.

```
[35]: # Combine generators into one which yields image and mask
train_generator = zip(image_generator, mask_generator)
```

```
[37]: def build_fcnn():
    inputs = Input((128, 128, 3))

    # Encoder
    conv1 = Conv2D(128, (3, 3), activation='relu', padding='same')(inputs)
    pool1 = MaxPooling2D((2, 2))(conv1)

    conv2 = Conv2D(256, (3, 3), activation='relu', padding='same')(pool1)
    pool2 = MaxPooling2D((2, 2))(conv2)

    # Decoder
    conv3 = Conv2D(256, (3, 3), activation='relu', padding='same')(pool2)
    up1 = Conv2DTranspose(128, (2, 2), strides=(2, 2), padding='same')(conv3)

    conv4 = Conv2D(128, (3, 3), activation='relu', padding='same')(up1)
    up2 = Conv2DTranspose(64, (2, 2), strides=(2, 2), padding='same')(conv4)

    outputs = Conv2D(1, (1, 1), activation='sigmoid', padding='same')(up2)

    model = Model(inputs, outputs)
    return model
```

```
[38]: model = build_fcnn()
model.compile(optimizer='adam', loss='binary_crossentropy',
             metrics=['accuracy'])
model.summary()
```

Model: "functional_5"

Layer (type) Param #	Output Shape	
input_layer_5 (InputLayer) 0	(None, 128, 128, 3)	
conv2d_6 (Conv2D) 3,584	(None, 128, 128, 128)	
max_pooling2d_6 (MaxPooling2D) 0	(None, 64, 64, 128)	
conv2d_7 (Conv2D) 295,168	(None, 64, 64, 256)	
max_pooling2d_7 (MaxPooling2D) 0	(None, 32, 32, 256)	
conv2d_8 (Conv2D) 590,080	(None, 32, 32, 256)	
conv2d_transpose (Conv2DTranspose) 131,200	(None, 64, 64, 128)	
conv2d_9 (Conv2D) 147,584	(None, 64, 64, 128)	
conv2d_transpose_1 (Conv2DTranspose) 32,832	(None, 128, 128, 64)	
conv2d_10 (Conv2D) 65	(None, 128, 128, 1)	

Total params: 1,200,513 (4.58 MB)

Trainable params: 1,200,513 (4.58 MB)

Non-trainable params: 0 (0.00 B)

```
[40]: # Train the FCNN model
def combined_generator(image_gen, mask_gen):
    while True: # Keep yielding data indefinitely
        img_batch = next(image_gen)
        mask_batch = next(mask_gen)
        yield img_batch, mask_batch # Keras expects (input, target)
# Fit the model with the custom generator
train_generator = combined_generator(image_generator, mask_generator)
model.fit(train_generator, steps_per_epoch=len(image_generator), epochs=5)
```

```
Epoch 1/5
3/3          6s 2s/step -
accuracy: 0.6881 - loss: 0.5807
Epoch 2/5
3/3          6s 2s/step -
accuracy: 0.6818 - loss: 0.5611
Epoch 3/5
3/3          6s 2s/step -
accuracy: 0.6803 - loss: 0.5453
Epoch 4/5
3/3          6s 2s/step -
accuracy: 0.6850 - loss: 0.5678
Epoch 5/5
3/3          6s 2s/step -
accuracy: 0.6869 - loss: 0.5435
```

[40]: <keras.src.callbacks.history.History at 0x1ccb90ed8d0>

```
[42]: # Sample image for prediction
sample_image = image_generator[0][0]
predicted_mask = model.predict(np.expand_dims(sample_image, axis=0))[0]
```

```
1/1          0s 169ms/step
```

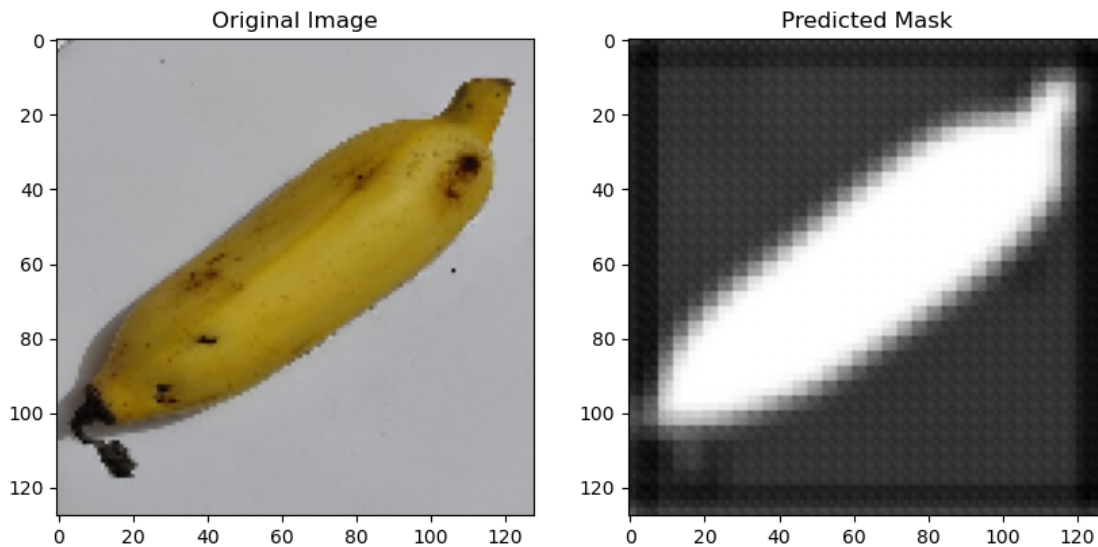
```
[43]: # Display the original image and predicted mask
plt.figure(figsize=(10, 5))

plt.subplot(1, 2, 1)
plt.title("Original Image")
plt.imshow(sample_image)

plt.subplot(1, 2, 2)
plt.title("Predicted Mask")
```

```
plt.imshow(predicted_mask.squeeze(), cmap='gray')

plt.show()
```



5 Exe4(Cityscapes dataset)

```
[62]: import glob
import numpy as np
import matplotlib.pyplot as plt
from PIL import Image
import tensorflow as tf
from tensorflow.keras.layers import Input, Conv2D, MaxPooling2D, Conv2DTranspose
from tensorflow.keras.models import Model
```

```
[63]: BASE_DIR = 'E:/126156040/cityscapes_data'
IMG_HEIGHT, IMG_WIDTH = 128, 128
NUM_CLASSES = 13
```

```
[64]: def load_data(path):
    X, Y = [], []
    bins = np.arange(20, 260, 20)
    for f in glob.glob(path + "/*.jpg"):
        img = Image.open(f)
        X.append(np.array(img.crop((0, 0, 256, 256)).convert('RGB')).
↪resize((IMG_WIDTH, IMG_HEIGHT))))
        msk = img.crop((256, 0, 512, 256)).convert('L').resize((IMG_WIDTH,
↪IMG_HEIGHT))
```

```

        Y.append(np.digitize(np.array(msk), bins))
    return np.array(X, np.float32) / 255.0, np.array(Y, np.int32)

```

```

[65]: def build_fcn(shape, classes):
        x = inputs = Input(shape)
        x = Conv2D(32, 3, activation='relu', padding='same')(x)
        x = MaxPooling2D()(x)
        x = Conv2D(64, 3, activation='relu', padding='same')(x)
        x = MaxPooling2D()(x)
        x = Conv2D(128, 3, activation='relu', padding='same')(x)
        x = Conv2DTranspose(64, 2, strides=2, padding='same')(x)
        x = Conv2DTranspose(256, 2, strides=2, padding='same')(x)
        outputs = Conv2D(classes, 1, activation='softmax')(x)
        return Model(inputs, outputs)

```

```

[66]: X_train, y_train = load_data(BASE_DIR + "/train")
        X_val, y_val = load_data(BASE_DIR + "/val")
        model = build_fcn((IMG_HEIGHT, IMG_WIDTH, 3), NUM_CLASSES)
        model.compile(optimizer='adam', loss='sparse_categorical_crossentropy',
            ↪metrics=['accuracy'])

```

```

[67]: model.fit(X_train, y_train, batch_size=8, epochs=2, validation_data=(X_val,
    ↪y_val))

        i = np.random.randint(len(X_val))
        pred = np.argmax(model.predict(X_val[i:i+1]), -1)[0]

```

```

Epoch 1/2
372/372          168s 445ms/step -
accuracy: 0.4154 - loss: 1.7124 - val_accuracy: 0.5852 - val_loss: 1.3322
Epoch 2/2
372/372          153s 412ms/step -
accuracy: 0.6278 - loss: 1.2027 - val_accuracy: 0.6484 - val_loss: 1.1691
1/1              0s 119ms/step

```

```

[68]: def show(title, img, cmap=None):
        plt.imshow(img, cmap=cmap)
        plt.title(title)
        plt.axis("off")
        plt.show()

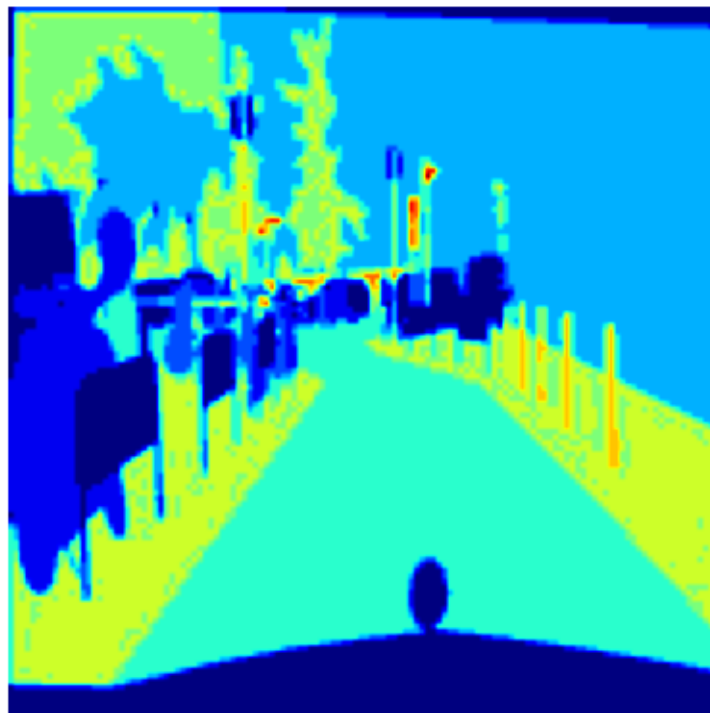
        show("Image", X_val[i])
        show("True Mask", y_val[i], cmap='jet')
        show("Predicted Mask", pred, cmap='jet')

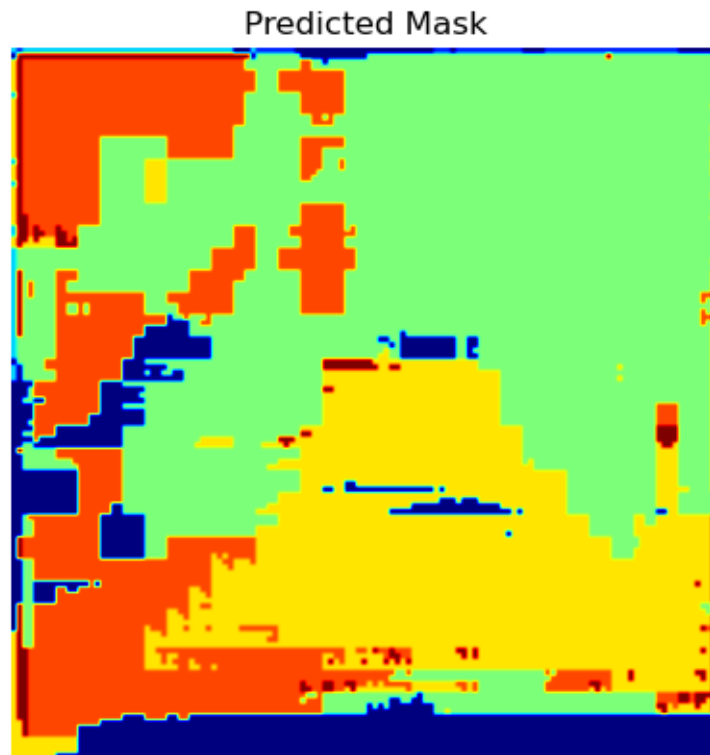
```


Image



True Mask





6 Exe 5

```
[44]: import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras.datasets import mnist
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Input, Dense
from tensorflow.keras.optimizers import Adam
```

```
[45]: (x_train, _), (x_test, _) = mnist.load_data()
x_train = x_train.astype('float32') / 255.
x_test = x_test.astype('float32') / 255.
x_train = x_train.reshape((len(x_train), np.prod(x_train.shape[1:])))
x_test = x_test.reshape((len(x_test), np.prod(x_test.shape[1:])))
```

```
[46]: input_dim = x_train.shape[1] # 784 (28*28)
encoding_dim = 32 # You can adjust this value
input_img = Input(shape=(input_dim,))
encoded = Dense(encoding_dim, activation='relu')(input_img)
```

```
decoded = Dense(input_dim, activation='sigmoid')(encoded)
autoencoder = Model(input_img, decoded)
encoder = Model(input_img, encoded)
autoencoder.compile(optimizer=Adam(), loss='binary_crossentropy')
```

```
[47]: autoencoder.fit(x_train, x_train,
epochs=50,
batch_size=256,
shuffle=True,
validation_data=(x_test, x_test))
```

```
Epoch 1/50
235/235          2s 4ms/step -
loss: 0.3852 - val_loss: 0.1861
Epoch 2/50
235/235          1s 3ms/step -
loss: 0.1768 - val_loss: 0.1522
Epoch 3/50
235/235          1s 3ms/step -
loss: 0.1482 - val_loss: 0.1327
Epoch 4/50
235/235          1s 3ms/step -
loss: 0.1309 - val_loss: 0.1202
Epoch 5/50
235/235          1s 3ms/step -
loss: 0.1195 - val_loss: 0.1120
Epoch 6/50
235/235          1s 3ms/step -
loss: 0.1117 - val_loss: 0.1060
Epoch 7/50
235/235          1s 3ms/step -
loss: 0.1062 - val_loss: 0.1018
Epoch 8/50
235/235          1s 3ms/step -
loss: 0.1024 - val_loss: 0.0988
Epoch 9/50
235/235          1s 2ms/step -
loss: 0.0996 - val_loss: 0.0965
Epoch 10/50
235/235          1s 3ms/step -
loss: 0.0976 - val_loss: 0.0951
Epoch 11/50
235/235          1s 3ms/step -
loss: 0.0958 - val_loss: 0.0941
Epoch 12/50
235/235          1s 3ms/step -
loss: 0.0954 - val_loss: 0.0936
```

Epoch 13/50
235/235 1s 3ms/step -
loss: 0.0946 - val_loss: 0.0931
Epoch 14/50
235/235 1s 3ms/step -
loss: 0.0943 - val_loss: 0.0929
Epoch 15/50
235/235 1s 3ms/step -
loss: 0.0941 - val_loss: 0.0928
Epoch 16/50
235/235 1s 3ms/step -
loss: 0.0939 - val_loss: 0.0925
Epoch 17/50
235/235 1s 3ms/step -
loss: 0.0936 - val_loss: 0.0924
Epoch 18/50
235/235 1s 3ms/step -
loss: 0.0936 - val_loss: 0.0922
Epoch 19/50
235/235 1s 3ms/step -
loss: 0.0936 - val_loss: 0.0922
Epoch 20/50
235/235 1s 3ms/step -
loss: 0.0935 - val_loss: 0.0921
Epoch 21/50
235/235 1s 3ms/step -
loss: 0.0934 - val_loss: 0.0921
Epoch 22/50
235/235 1s 3ms/step -
loss: 0.0932 - val_loss: 0.0920
Epoch 23/50
235/235 1s 3ms/step -
loss: 0.0933 - val_loss: 0.0920
Epoch 24/50
235/235 1s 3ms/step -
loss: 0.0931 - val_loss: 0.0919
Epoch 25/50
235/235 1s 3ms/step -
loss: 0.0930 - val_loss: 0.0918
Epoch 26/50
235/235 1s 3ms/step -
loss: 0.0931 - val_loss: 0.0918
Epoch 27/50
235/235 1s 3ms/step -
loss: 0.0930 - val_loss: 0.0918
Epoch 28/50
235/235 1s 3ms/step -
loss: 0.0928 - val_loss: 0.0918

Epoch 29/50
235/235 1s 2ms/step -
loss: 0.0928 - val_loss: 0.0917
Epoch 30/50
235/235 1s 3ms/step -
loss: 0.0930 - val_loss: 0.0917
Epoch 31/50
235/235 1s 3ms/step -
loss: 0.0928 - val_loss: 0.0916
Epoch 32/50
235/235 1s 3ms/step -
loss: 0.0928 - val_loss: 0.0917
Epoch 33/50
235/235 1s 3ms/step -
loss: 0.0927 - val_loss: 0.0916
Epoch 34/50
235/235 1s 3ms/step -
loss: 0.0929 - val_loss: 0.0916
Epoch 35/50
235/235 1s 3ms/step -
loss: 0.0929 - val_loss: 0.0916
Epoch 36/50
235/235 1s 3ms/step -
loss: 0.0928 - val_loss: 0.0916
Epoch 37/50
235/235 1s 3ms/step -
loss: 0.0928 - val_loss: 0.0916
Epoch 38/50
235/235 1s 3ms/step -
loss: 0.0927 - val_loss: 0.0915
Epoch 39/50
235/235 1s 3ms/step -
loss: 0.0926 - val_loss: 0.0915
Epoch 40/50
235/235 1s 3ms/step -
loss: 0.0927 - val_loss: 0.0915
Epoch 41/50
235/235 1s 3ms/step -
loss: 0.0927 - val_loss: 0.0915
Epoch 42/50
235/235 1s 3ms/step -
loss: 0.0926 - val_loss: 0.0915
Epoch 43/50
235/235 1s 3ms/step -
loss: 0.0923 - val_loss: 0.0916
Epoch 44/50
235/235 1s 3ms/step -
loss: 0.0926 - val_loss: 0.0915

```

Epoch 45/50
235/235          1s 3ms/step -
loss: 0.0927 - val_loss: 0.0915
Epoch 46/50
235/235          1s 3ms/step -
loss: 0.0925 - val_loss: 0.0914
Epoch 47/50
235/235          1s 2ms/step -
loss: 0.0925 - val_loss: 0.0915
Epoch 48/50
235/235          1s 3ms/step -
loss: 0.0924 - val_loss: 0.0915
Epoch 49/50
235/235          1s 3ms/step -
loss: 0.0926 - val_loss: 0.0915
Epoch 50/50
235/235          1s 3ms/step -
loss: 0.0926 - val_loss: 0.0914

```

[47]: <keras.src.callbacks.history.History at 0x1ccc540ba50>

```

[48]: encoded_imgs = encoder.predict(x_test)
      decoded_imgs = autoencoder.predict(x_test)

```

```

313/313          0s 589us/step
313/313          0s 652us/step

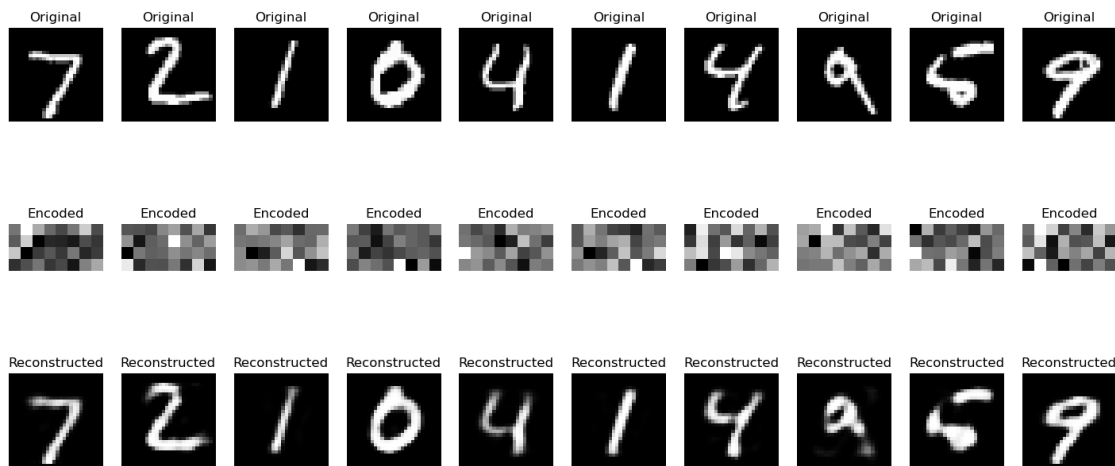
```

```

[51]: n = 10 # Number of images to display
      plt.figure(figsize=(18, 8))
      for i in range(n):
          # Original images
          ax = plt.subplot(3, n, i + 1)
          plt.imshow(x_test[i].reshape(28, 28))
          plt.title("Original")
          plt.gray()
          ax.axis('off')
          # Encoded images (dimensionality reduced representation)
          ax = plt.subplot(3, n, i + 1 + n)
          plt.imshow(encoded_imgs[i].reshape(4, 8)) # Reshape for visualization
          plt.title("Encoded")
          plt.gray()
          ax.axis('off')
          # Reconstructed images
          ax = plt.subplot(3, n, i + 1 + 2 * n)
          plt.imshow(decoded_imgs[i].reshape(28, 28))
          plt.title("Reconstructed")
          plt.gray()
          ax.axis('off')

```

```
plt.show()
```



```
[ ]:
```